

JAEBUM PARK

Graduate Student Researcher & Ph.D. Student

Electrical and Computer Engineering Department at University of California, San Diego (UCSD)

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RESEARCH INTERESTS

Wireless Digital Communications, Mobile Communications, Physical Layer Algorithm

- Massive MIMO Techniques and Modem Algorithms toward 6G Candidate Frequency Bands
- Next generation RAN architectures (Open RAN, virtualized RAN)
- AI/ML-based digital predistortion (DPD)
- Energy-efficient RAN
- 3GPP RAN1 Standards (4G LTE, 5G NR)

EDUCATION

University of California, San Diego (UCSD)

Ph.D. Student in Electrical & Computer Engineering

- Wireless Systems Innovation Laboratory (WSIL)
- Advisor: *Prof. Robert W. Heath, Jr.*

San Diego, CA, USA
Sep. 2024 - Present

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in Electrical & Electronic Engineering

- Thesis: "Reduced Feedback Beamforming with Large Antenna Array in Wireless Backhaul Downlink System"
- Laboratory for Information Transmission, KAIST
- Advisor: *Prof. Hyuncheol Park*

Daejeon, S. Korea
Feb. 2017

Yonsei University

B.S. in Electrical & Electronic Engineering

- Completed mandatory military service during this period (Mar. 2010 – Jan. 2012)
- Received A+ grades in all mathematics-related subjects

Seoul, S. Korea
Feb. 2015

PUBLICATIONS [Journal]

Jaebum Park and Robert W. Heath, Jr., "Two-Timescale Receive Combining for Fronthaul-Constrained Uplink XL-MIMO with Visibility Regions," in preparation for submission to *IEEE Transactions on Wireless Communications*, 2027.

Jaebum Park, Yutao Xie, Amitava Ghosh, Yuanyuan Shi, and Robert W. Heath, Jr., "Carbon-Aware Joint Resource and Energy Scheduling for Cellular Base Stations," in preparation for submission to *IEEE Communications Letters*, 2026.

Seunghun Lee, **Jaebum Park**, Hyuncheol Park, and Robert W. Heath, Jr., "Distributed Graph Neural Network for Scalable Beam Configuration in Cell-Free MIMO Networks," in preparation for submission to *IEEE Transactions on Wireless Communications*, 2026.

Jaebum Park, Chan-Byoung Chae, and Robert W. Heath, Jr., "Energy Efficiency Optimization in Distributed MIMO vRAN via Cross-Layer Link Abstraction," *IEEE Transactions on Wireless Communications*, under revision, 2026.

Boyu Ning, Yike Xie, **Jaebum Park**, Songjie Yang, Weidong Mei, Emil Björnson, and Robert W. Heath, Jr., "Broadcasting in Centimeter-Wave G-MIMO Communication Systems: Beam Shaping is All You Need," *IEEE Transactions on Wireless Communications*, under revision, 2026.

Jaebum Park, Chan-Byoung Chae, and Robert W. Heath, Jr., "Accelerating vRAN and O-RAN with SIMD: Architectural Perspectives and Performance Evaluation," *IEEE Communications Magazine*, vol. 64, no. 11, 2026.

Boyu Ning, Haifan Yin, Sixu Liu, Hao Deng, Songjie Yang, Yuchen Zhang, Weidong Mei, David Gesbert, **Jaebum Park**, Robert W. Heath, Jr., and Emil Björnson, "Precoding Matrix Indicator in the 5G NR Protocol: A Tutorial on 3GPP Beamforming Codebooks," *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 4581-4623, 2026.

Jaebum Park, Girim Kwon, and Hyuncheol Park, "Reduced Feedback Beamforming with a Large Antenna Array in Wireless Backhaul Downlink Systems," 2016 The Journal of Korean Institute of Communications and Information Sciences (J-KICS), 41(12), 1903-1913.

PUBLICATIONS [Conference]

Jaebum Park, Chan-Byoung Chae, and Robert W. Heath, Jr., "Energy Efficiency Optimization for 5G NR PDSCH: A Cross-Layer Link Abstraction Framework," submitted to *IEEE ICC*, Washington DC, USA, 2027.

Jaebum Park and Robert W. Heath, Jr., "Fronthaul-constrained low-rank RX combiner design for uplink XL-MIMO," *2026 60th Asilomar Conference on Signals, Systems, and Computers*, Pacific Grove, CA, USA, 2026.

Jaebum Park, Chan-Byoung Chae, and Robert W. Heath, Jr., "Switch-DFT: Adaptive Waveform and MIMO Switching for Energy-Efficient Base Stations," in *Proc. 2026 IEEE ICC Workshops*, Glasgow, UK, 2026, pp. 1-6.

Hyungsik Han, **Jaebum Park**, Jungmin Kim, Kitaek Bae, and Ilju Na, "Baseband Phase Noise Modeling and Analysis for 140 GHz THz DFT-s-OFDM System," in *Proc. 2022 IEEE Globecom Workshops*, Rio de Janeiro, Brazil, 2022, pp. 1748-1753.

Girim Kwon, **Jaebum Park**, and Hyuncheol Park, "A Two-stage Beamforming Scheme using Location Information in Massive MIMO Wireless Backhaul Channel," in *Proc. Symposium of the Korean Institute of Communications and Information Sciences*, 2017, 536-536.

Jaebum Park, Girim Kwon, and Hyuncheol Park, "A Study on Antenna Structures for Beamforming in Massive MIMO Wireless Backhaul Downlink System," in *Proc. Symposium of the Korean Institute of Communications and Information Sciences*, 2016, 1495-1496.

Girim Kwon, **Jaebum Park**, and Hyuncheol Park, "Time Resource Allocation of Time Division Transmission Scheme in Massive MIMO Wireless Backhaul Downlink System," in *Proc. Symposium of the Korean Institute of Communications and Information Sciences*, 2016, 52-53.

PATENTS

Jaebum Park, Youngjoon Kim, Sunho Park, Jiyeong Yang, Chanuk Jeong, "Apparatus and Method for Performing Port Combining in Wireless Communication System," KR/US patent, Application Number: 10-2023-0181792, Dec. 14, 2023 (U.S. counterpart application pending)

Youngjoon Kim, Daejung Yoon, **Jaebum Park**, Jiyeong Yang, Chanuk Jeong, "Method and Apparatus Using SVD-based Port Combining Signaling for O-DU," KR/US patent, Application Number: 10-2023-0178721, Dec. 11, 2023 (U.S. counterpart application pending)

Jaeha Lee, Taehyeong Kim, Hyunjung Nam, **Jaebum Park**, Joonah Park, "Electronic Apparatus and Controlling Method Thereof," US patent, Publication Number: 12094129, Sep. 17, 2024

Jaeyong Ju, Hyeran Lee, Hyunjung Nam, Miyoung Kim, **Jaebum Park**, Joonah Park, "Mobile Robot and Driving Method Thereof," US patent, Publication Number: 11801602, Oct. 31, 2023

EXPERIENCE AND PROJECTS

Center for Wireless Communications (CWC), UCSD

- 1) Graduate Student Researcher (GSR)
 - Developing energy-efficient MIMO algorithms for next generation RAN
 - AI/ML-based digital predistortion (DPD) technique
- 2) UCSD Research Expo 2025 / 2026
 - Presented a poster on “Accelerating MIMO algorithms in vRAN/O-RAN architectures” in 2025
 - Presented a poster on “3D spatio-temporal traffic-aware energy efficiency optimization for distributed MIMO systems” in 2026
- 3) UCSD CWC Research Review 2025 (05/30/2025)
 - Gave a poster presentation on accelerating MIMO algorithms in the vRAN/O-RAN architecture
- 4) San Diego Wireless Summit 2026
 - Presented a poster on SIMD-based acceleration for vRAN and O-RAN

San Diego, CA, USA
Sep. 2024 - Present

Samsung Research (SR), Samsung Electronics

Engineer (2017 – 2022) / Staff Engineer (2023 - 2024)

- 1) Task on Uplink Performance Improvement (ULPI) in O-RAN
 - Developed an advanced MIMO receiver to optimize the O-RAN functional split at option 7-2x
 - Proposed a low-complexity SVD-based compression method
- 2) 6G Testbed (TB) Development Project
 - Demonstrate a 100 Gbps transmission at the Samsung Tech Fair 2022
 - Defined 6G THz TB Key Performance Indicators (KPIs) & Specification (PDSCH, PUSCH, PDCCH, PUCCH, SSB, PRACH)
 - Created a Link-Level Simulator (LLS) for 6G TB (Floating/Fixed Point Modeling)
 - Implemented RTL projects for 6G TB hardware FPGA Platform
 - Developed MIMO receiver algorithms toward 6G candidate frequency bands (e.g., sub-THz, FR3)
- 3) AI-based Smart Machine Development Project
 - Improved the accuracy of lane detection algorithm for autonomous driving by using deep learning, such as convolutional neural network (CNN)
 - Created the human following algorithm for healthcare robots by combining various kinds of sensor data
 - Showcased the Samsung Bot at 2019 CES, Las Vegas, U.S.

Seoul, S. Korea
Jan. 2017 - Jul. 2024

Korea Advanced Institute of Science and Technology (KAIST)

- 1) Research Assistant (RA)
 - Performed modelling of a ray-tracing based clustered delay line (CDL) channel
 - Designed beamforming matrices based on reduced feedback
 - Wireless Backhaul Downlink Systems Project, sponsored by ETRI
 - Advisor: *Prof. Hyuncheol Park*
- 2) Teaching Assistant (TA)
 - Served as a TA for undergraduate experimental subjects, such as C++ Programming and Electronic & Electrical Circuit Design
 - Provided mentoring to undergraduate students

Daejeon, S. Korea
Jan. 2015 - Dec. 2016

Yonsei University

Undergraduate Internship Student

- Communications & Network Laboratories, Yonsei University
- Advisor: *Prof. Jong-Moon, Chung*

Seoul, S. Korea
Jan. 2014 – Jul. 2014

AWARDS AND SCHOLARSHIPS

IEEE ICC 2026 Travel Grant

- NSF Student Travel Grant for ICC 2026

2026.05

National Scholarships

- Tuition exemption and full scholarship during master's program

2015 - 2017

Honor Scholarships

- 4 scholarships for academic excellence out of 8 undergraduate semesters

2008 - 2015

Undergraduate Thesis Award

- Creative Engineering Scholarship
- Award for the most outstanding undergraduate thesis

2014

1st Award of Mathematical Contest

- 1st prize out of 300 students

2012

Dean's List Award

- Honors Program for the most outstanding student
- GPA of 4.42/4.5 (All A+ grades except for one A)

2012

COMPUTER SKILLS

MATLAB, C/C++, Python, Verilog, RTL, Github, Linux OS, etc.

HOBBIES AND PERSONAL INTERESTS

I have been an avid ice hockey player for around 20 years.
Additionally, I enjoy playing golf, soccer, and other sports.